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An Inverse Finite Element *u*/*p*-Formulation to Predict the Unloaded State of *In Vivo* Biological Soft Tissues

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Abstract—Physically realistic patient-specific biomechanical modelling is of paramount importance for many medical applications, where the geometry of tissues or organs is usually constructed from *in vivo* images. However, it is common for such biological structures to correspond to a deformed state due to being under external loadings. This necessitates the determination of the stress distribution of the known deformed state through an inverse analysis approach. To achieve this, we propose here a generalised finite element displacement/pressure (*u*/*p*)-formulation for evaluating the unloaded configuration of *in vivo* biological soft tissues that exhibit quasi-incompressible behaviour under finite deformations. Validity and applicability of the proposed numerical framework to practical inverse analysis problems in biomechanics is demonstrated through various numerical examples. The corresponding simulations utilise *in vivo* measurements of patient-specific geometries derived from different medical imaging modalities, and include recovery of the pressure-free configuration of human aortas and the gravity-free shape of the female breast.

Keywords—Inverse analysis, Incompressible, FEM, Tissue deformation, Initial stress, Patient-specific modelling.

INTRODUCTION

In computational continuum solid mechanics, the forward (or direct) problem involves the computation of the strain and stress field of a given elastic body with known prescribed initial and boundary conditions, and specific mechanical properties. Hence, in both linear and nonlinear analysis, the final (or deformed) configuration of the elastic body can be computed at the end of the numerical simulation providing the solution of the forward problem. However, there is also a class of problems where the deformed configuration of the elastic body is known, and its undeformed state is to be evaluated. These problems are referred in the literature as inverse problems,[1,13] where one seeks to recover the stress-free or unloaded state of a structure given its current deformed state, the boundary conditions and the mechanical properties of the solid.

In particular, when modelling bioengineering problems, one often encounters simulation of biological structures which, in the observed state, are loaded by external forces. More specifically, patient-specific geometries are usually acquired *in vivo* through medical imaging (e.g., magnetic resonance imaging (MRI), computed tomography scanning, 3D photography, *etc*.).However, it is generally not possible, using these modalities, to measure tissue pre-stressing or accurately quantify the level of tissue deformation from the unloaded state. Therefore, for high numerical simulation accuracy, it is important for the analyst to incorporate into the biomechanical model all necessary information including any pre-stressing of the solid structure. For example, as reported by recent cardiovascular studies[25,38] directly incorporating the original scanned geometry of an artery/vein into fluid-structure interaction simulations can give inaccurate predictions of the haemodynamics and the vascular biomechanical response. In addition, for pure structural analysis of a blood vessel wall at the full physiological pressure load, incorrect wall distensibility and stress distributions will be predicted if erroneous initial unstressed conditions are assumed. Very recently, inverse analysis approaches have been used for the estimation of the

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wall distensibility of aortic aneurysms to aid reliable patient-specific rupture risk assessment.[26] Also, inverse algorithms have been utilised to compute the level of pre-stress of *in vivo* blood vessel walls of patient-specific models.[6] Similar issues, in various forms, are common to all applications of biomechanical modelling based on *in vivo* medical imaging.

In the disparate field of cancer research, MRI is recommended for breast cancer patients in the UK if mammography is inadequate for lesion assessment, disagrees with ultrasound on the extent of the disease, or if invasive lobular cancer is present and breast conserving surgery is being considered.[28] In clinical practice, the use of pre-operative MRI varies greatly[30] but is frequently available prior to surgery[29] and has increased substantially since 2000 in the USA in particular.[18] The utility of routinely acquired MRI specifically for image guided surgical planning (i.e., for verification of tumour size, location and precise estimation of tumour resection margins) is not currently exploited, primarily we believe, due to the discrepancy between the prone pose of the patient during MRI acquisition and the supine configuration adopted during surgery. A number of publications have proposed computational methodologies for breast deformation simulation in mammography compression, and breast reduction or augmentation interventions but—as reported in the following cited papers—are dependent upon an accurate estimation of the gravity-free configuration of the patient-specific breast model.

In this and the following paragraph we will use these exemplary biomechanical applications to examine, in more detail, the methods developed to derive the stress-free configuration of patient-specific models. To predict vessel wall stress distribution for true diastolic vessel geometries at a given pressure loading, de Putter *et al.*[7] developed a methodology using a backward incremental approach. Their method is based on the conventional updated Lagrangian finite element (FE) framework, where a backward application of the computed forward deformations is used. In the same year, Lu *et al.*[20] presented an inverse computational methodology, founded on the developments of Govindjee and Mihalic,[14] for predicting pre-stress loadings and the reference state of finitely deforming anisotropic hyperelastic solids. Raghavan and colleagues[32] presented an iterative based algorithm to predict the pressure-free shape of patient-specific abdominal aortic aneurysm (AAA) geometries using isotropic hyperelastic models. Their work was later enhanced by Riveros *et al.*[35] who also considered the anisotropic biomechanical behavior of the aortic wall. Moreover, Gee and his colleagues[11,12] proposed a modified updated Lagrangian formulation, similar in concept to the approach of de Putter,[7] to predict the stress-free configuration of blood vessels with a particular focus on patient-specific AAA geometries as 3D solid structures. On the other hand, Lu *et al.*[22] proposed an inverse method for thin-wall structures modelled as geometrically exact stress resultant shells. Their numerical approach was subsequently utilised for the stress analysis of patient-specific cerebral aneurysms.[43] More recently, Bols *et al.*[2] formulated an iterative backward displacement method to recover the zero-pressure configuration of human blood vessel geometries and predict the *in vivo* stress distribution through a fixed-point algorithm.

From the breast image registration perspective and inverse biomechanical modelling, Carter *et al.*[3] proposed an iterative FE methodology which was then improved by Eiben *et al.*[9] to evaluate the unloaded shape of patient-specific breast models. The algorithm for recovering the zero-gravity breast shape starts with an initial guess of the unloaded configuration by inverting the direction of gravity. Subsequently, the prone loaded configuration is loaded with gravity in the posterior direction (i.e., the direction is reversed) and the estimated reference configuration is loaded in the opposite direction, until the nodal distance between the two forward simulations satisfy a convergence tolerance. More recently, they compared the performance of their methodology with a standard inverse design approach in prone-to-supine image registration for various patient-specific breast geometries, and assessed the material parameter influence on the sensitivity of the zero-gravity numerical predictions.[10] Rajagopal and colleagues[34] presented a pure Lagrangian iterative scheme that numerically evaluates the unloaded configuration of a solid body through a standard forward finite deformation approach. To achieve this, an estimated undeformed state is perturbed and the unbalanced forces residual is evaluated repeatedly until convergence is achieved. However, despite the ease to implement the aforementioned methodologies in existing forward analysis FE codes, their ability to accurately recover the reference state relies heavily on the initial estimate of the zero-gravity configuration of the breast geometry. Additionally, the level of sub-incrementation—or the magnitude of the external perturbation applied[33]—is reported to play an important role on the stability of the inverse algorithms, while the convergence performance is not optimal. Very recently, Joldes and colleagues[17] proposed an interesting approach that utilises the re-parametrisation of the linear momentum equations—conceptually based on the work of Govindjee and Mihalic[13]—which were discretised using an explicit total Lagrangian framework and subsequently implemented using GPU technology.

In this paper we propose an accurate and robust FE framework for the inverse analysis of nearly incompressible hyperelastic materials, with special emphasis on the simulation of biological soft tissues and organs. The methodology is based on an Eulerian displacement/pressure (u/p)-formulation, similar in concept to the Lagrangian two-field forward analysis approach of Sussman and Bathe[37] and Yamada and Kikuchi.[42] The proposed full-implicit inverse algorithm can achieve very fast convergence—in comparison to explicit codes[17]—while simultaneously being very efficient in enforcing material incompressibility, as opposed to pure displacement based inverse formulations.[2,9,12] The proposed methodology is founded on the concept of re-parametrisation of the balance equation in the current (deformed) setting.[14] The numerical procedure assumes a non-specific constitutive description of the isochoric term, seamlessly incorporates anisotropic constitutive models, and circumvents the simplification assumptions made in previous works of discontinuous pressure across the element boundaries.[20,21]

MATERIALS AND METHODS

Let Ω be the analysed domain of a biological structure which is assumed as a continuous elastic body, surrounded by boundary Γ, in its reference (undeformed) placement. In finite deformation analysis, a material point $\mathbf{X} \in \Omega$ can be mapped to the current (deformed) configuration, ω, by the motion: $\mathbf{x} = \Phi(\mathbf{X})$, where $\mathbf{x} \in \omega$.In this regard, the deformation gradient tensor, defined by $\mathbf{F} = \partial\mathbf{x}/\partial\mathbf{X}$, can be used to measure deformation from Ω to ω.Now, assume a hyperelastic material with a potential function $\bar{W}$ representing energy per unit volume of the reference configuration. The Eulerian stress measure, namely the Cauchy stress tensor, can be expressed with respect to the stored-energy function $\bar{W}$[23]—also referred in the literature as strain-energy or free-energy function—as

$$\boldsymbol{\sigma} = 2J^{-1}\,\mathbf{B}\cdot\frac{\partial\bar{W}}{\partial\mathbf{B}}, \tag{1}$$

where J is the determinant of the deformation gradient tensor, and $\mathbf{B} = \mathbf{F}\cdot\mathbf{F}^T$ is the Finger tensor. In this study, we describe biological soft tissues biomechanical response as Green-elastic quasi-incompressible, undergoing finite deformations where any rate-dependency effects are ignored.

In the inverse problem addressed here, the aim is to evaluate the initial (undeformed) configuration of the elastic body given its deformed state. Thus, the inverse problem can be formulated based on the inverse motion: $\mathbf{X} = \boldsymbol{\phi}(\mathbf{x})$, where the mappings are given by definition through $\Phi = \boldsymbol{\phi}^{-1}$, and by re-parametrisation of the governing equations to the inverse setting.[13] This means that re-parametrisation is achieved through the inverse deformation gradient tensor $\mathbf{f} = \partial\mathbf{X}/\partial\mathbf{x}$, and the constitutive relationship is described using the inverse deformation $\mathbf{c} = \mathbf{f}^T\cdot\mathbf{f}$, where $\mathbf{c} = \mathbf{B}^{-1}$. However, given that the forward stored-energy function $\bar{W}$ for nearly incompressible hyperelastic solids is conveniently described through the modified invariants: $J_1 = I_1(\mathbf{B})\,J^{-2/3}$, $J_2 = I_2(\mathbf{B})\,J^{-4/3}$ and J, the description of the corresponding stored-energy function $\bar{w}$ in the inverse setting can be given with respect to the inverse modified invariants:[20] $\jmath_1 = J_2$, $\jmath_2 = J_1$ and $\jmath = J^{-1}$. Hence, invoking the above relations and substituting into Eq. (1) one can obtain the Cauchy stress tensor in the inverse setting (analytic derivation is provided in the Appendix)

$$\boldsymbol{\sigma} = -2\jmath\,\frac{\partial\bar{w}}{\partial\mathbf{c}}\cdot\mathbf{c}. \tag{2}$$

The present two-field mixed formulation considers separate interpolation for the displacement and hydrostatic pressure state variables. However, in this work, we propose a generalised FE methodology to solve inverse finite deformation elastostatics of near-incompressible and incompressible solids. In contrast to the papers of Govindjee and Mihalic[14] and Lu *et al.*,[20] the present formulation is not restricted by the assumption of constant pressure per element (i.e., discontinuous across elements' interface).

In common with previous theoretical developments,[37,42] we assume a modified potential $\bar{W} + Q$, where Q is a potential function of the (separately interpolated) pressure variable p. The principle of virtual work with respect to the deformed (known) configuration reads

$$\mathcal{G}_{\mathrm{U}}[\boldsymbol{\phi}(\mathbf{u}), p] = \int_{\omega} \delta(\bar{W} + Q)\,\mathrm{d}V - \mathcal{R} = 0, \tag{3}$$

where $\mathcal{R}$ is the external virtual work. In addition, the modified potential with respect to the pressure variable must also satisfy the following constraint equation

$$\mathcal{G}_{\mathbf{P}}[\boldsymbol{\phi}(\mathbf{u}), p] = \int_{\omega} \frac{\bar{p} - p}{P(\bar{p})}\,\delta p\,\mathrm{d}V = 0, \tag{4}$$

where δp is a variation of the independent pressure variable, while $\bar{p}$ denotes the hydrostatic pressure computed directly from the displacements, which is subsequently given by the expression: $\bar{p} = -\partial\bar{W}/\partial J$. The scalar operator function P of $\bar{p}$ is defined as $\partial\bar{p}/\partial J$ while—following Sussman and Bathe[37]—the potential quantity Q is given by

$$Q = -\frac{(\bar{p} - p)^2}{2P(\bar{p})}. \tag{5}$$

Assuming proper discretisation of Eqs. (3) and (4), and differentiating with respect to the corresponding state variables (displacement vector and hydrostatic pressure), the following system of the governing equations for a finite element in incremental form can be derived:

$$\begin{bmatrix} \mathbf{K}_{\mathrm{UU}} & \mathbf{K}_{\mathrm{UP}} \\ \mathbf{K}_{\mathrm{PU}} & \mathbf{K}_{\mathrm{PP}} \end{bmatrix} \cdot \begin{Bmatrix} \Delta\mathbf{u} \\ \Delta p \end{Bmatrix} = \begin{Bmatrix} \mathbf{R}-\mathbf{F}_{\mathrm{U}} \\ -\mathbf{F}_{\mathrm{P}} \end{Bmatrix}, \tag{6}$$

where the left-hand-side array corresponds to the Jacobian matrix and the right-hand-side vector to the residual vector, whose evaluation is required in iterative solution strategies (e.g., the Newton–Raphson scheme) and $\Delta\mathbf{u}$, Δp the nodal displacement and pressure increments. The vector quantities of the above system, at an element level, are given by

$$\mathbf{R} = \int_{\omega^e} \mathbf{N}_{\mathrm{U}}^{(i)} \cdot \mathbf{b}\,\mathrm{d}V + \int_{\gamma_T^e} \mathbf{N}_{\mathrm{U}}^{(i)} \cdot \bar{\mathbf{t}}\,dS, \tag{7a}$$

$$\mathbf{F}_{\mathrm{U}} = \int_{\omega^e} \mathbf{B}_{\mathrm{L}}^{(i)T} \cdot \tilde{\boldsymbol{\sigma}}\,\mathrm{d}V, \tag{7b}$$

$$\mathbf{F}_{\mathrm{P}} = \int_{\omega^e} \mathbf{N}_{\mathrm{P}}^{(i)} \frac{\bar{p}-p}{P(\bar{p})}\,\mathrm{d}V, \tag{7c}$$

where $\mathbf{b}$ is the body force vector, and $\bar{\mathbf{t}}$ the traction vector prescribed on the boundary γ_T of the deformed body. $\mathbf{B}_{\mathrm{L}}$ is the deformation matrix (as in the conventional small-deformation forward analysis case) that contains spatial derivatives of the *i*-node displacement-base shape function. $\mathbf{N}_{\mathrm{U}}$ and $\mathbf{N}_{\mathrm{P}}$ are diagonal arrays whose components are the corresponding shape functions of the displacement and pressure degrees of freedom respectively, while $\tilde{\boldsymbol{\sigma}}$ is the Voigt form of the stress tensor (see derivation in the Appendix):

$$\boldsymbol{\sigma} = \boldsymbol{\sigma} - \frac{\bar{p}-p}{P(\bar{p})}\frac{\partial\bar{p}}{\partial J}\,\mathbf{I} + \frac{(\bar{p}-p)^2}{2P^2(\bar{p})}\frac{\partial P(\bar{p})}{\partial J}\,\mathbf{I}, \tag{8}$$

with $\boldsymbol{\sigma}$ given in Eq. (2) and $\mathbf{I}$ is the identity matrix. The corresponding sub-matrices of the Jacobian are as follows:

$$\mathbf{K}_{\mathrm{UU}}^{(i,j)} = \int_{\omega^e} \mathbf{B}_{\mathrm{L}}^{(i)T} \cdot \tilde{\mathbf{C}}_{\mathrm{UU}} \cdot \mathbf{B}_{\mathrm{L}}^{(j)} \cdot \bar{\mathbf{f}}^T\,\mathrm{d}V, \tag{9a}$$

$$\mathbf{K}_{\mathrm{UP}}^{(i,j)} = \int_{\omega^e} \mathbf{B}_{\mathrm{L}}^{(i)T} \cdot \tilde{\mathbf{C}}_{\mathrm{UP}} \cdot \mathbf{N}_{\mathrm{P}}^{(j)}\,\mathrm{d}V, \tag{9b}$$

$$\mathbf{K}_{\mathrm{PU}}^{(i,j)} = \int_{\omega^e} \mathbf{N}_{\mathrm{P}}^{(i)} \cdot \tilde{\mathbf{C}}_{\mathrm{PU}} \cdot \mathbf{B}_{\mathrm{L}}^{(j)} \cdot \bar{\mathbf{f}}^T\,\mathrm{d}V, \tag{9c}$$

$$\mathbf{K}_{\mathrm{PP}}^{(i,j)} = \int_{\omega^e} \mathbf{N}_{\mathrm{P}}^{(i)} \cdot (C_{\mathrm{PP}}\,\mathbf{I}) \cdot \mathbf{N}_{\mathrm{P}}^{(j)}\,\mathrm{d}V, \tag{9d}$$

where $\bar{\mathbf{f}}$ is a square array whose diagonal components contain the inverse deformation gradient 3×3 matrix and the off-diagonal contains a 3×3 zero matrix.The tangent stiffness matrix $\tilde{\mathbf{C}}_{\mathrm{UU}}$ and the vectors $\tilde{\mathbf{C}}_{\mathrm{UP}}$, $\tilde{\mathbf{C}}_{\mathrm{PU}}$ are the Voigt form arrays of the fourth-order tensor C_{UU}, and second-order tensors $\mathbf{C}_{\mathrm{UP}}$ and $\mathbf{C}_{\mathrm{PU}}$ respectively, and the scalar quantity C_{PP} is given by the following analytic expressions

$$\begin{aligned}\mathsf{C}_{\mathrm{UU}} = \bar{\mathsf{C}}_{\mathrm{UU}} &- \frac{2}{P}\frac{\partial\bar{p}}{\partial J}\,\mathbf{I}\otimes\frac{\partial\bar{p}}{\partial\mathbf{c}} - \frac{2(\bar{p}-p)}{P}\,\mathbf{I}\otimes\frac{\partial^2\bar{p}}{\partial J\partial\mathbf{c}} \\ &+ \frac{2(\bar{p}-p)}{P^2}\frac{\partial\bar{p}}{\partial J}\,\mathbf{I}\otimes\frac{\partial P}{\partial\mathbf{c}} \\ &+ \frac{2(\bar{p}-p)}{P^2}\frac{\partial P}{\partial J}\,\mathbf{I}\otimes\frac{\partial\bar{p}}{\partial\mathbf{c}} + \frac{(\bar{p}-p)^2}{P^2}\,\mathbf{I}\otimes\frac{\partial^2 P}{\partial J\partial\mathbf{c}} \\ &- \frac{2(\bar{p}-p)^2}{P^3}\frac{\partial P}{\partial J}\,\mathbf{I}\otimes\frac{\partial P}{\partial\mathbf{c}},\end{aligned} \tag{10a}$$

$$\mathbf{C}_{\mathrm{UP}} = \frac{1}{P}\frac{\partial\bar{p}}{\partial J}\,\mathbf{I} - \frac{\bar{p}-p}{P^2}\frac{\partial P}{\partial J}\,\mathbf{I}, \tag{10b}$$

$$\mathbf{C}_{\mathrm{PU}} = \frac{2}{P}\frac{\partial\bar{p}}{\partial\mathbf{c}} - \frac{2(\bar{p}-p)}{P^2}\frac{\partial P}{\partial\mathbf{c}}, \tag{10c}$$

$$C_{\mathrm{PP}} = -P^{-1}, \tag{10d}$$

where $\bar{\mathsf{C}}_{\mathrm{UU}} \equiv 2\,\partial\boldsymbol{\sigma}/\partial\mathbf{c}$, $\partial\bullet/\partial\mathbf{c} = -(\jmath/2)[\partial\bullet/\partial J]\,\mathbf{c}^{-1}$, and the binary operator "$\otimes$" denotes tensor product. Analytical derivation of the first term in Eq. (10b) is given in the Appendix.

From Eq. (9b) it is evident that there is no geometric tangent stiffness involved in the inverse formulation, as opposed to the forward analysis case.[37] In addition, the material tangent stiffness $\partial\boldsymbol{\sigma}/\partial\mathbf{c}$ has only minor symmetries,[13] while the overall stiffness matrix is not symmetric since $\mathbf{K}_{\mathrm{UP}} \neq \mathbf{K}_{\mathrm{PU}}^T$. Apparently, if it is assumed that hydrostatic pressure is discontinuous across elements then the FE formulation of Govindjee and Mihalic[14] can be deduced by static condensation of the pressure degrees of freedom at the element level similarly to the forward analysis approach.[37] The inverse FE framework has been implemented in scalable *C++* code using the numerical libraries *PETSc* (http://www.mcs.anl.gov/petsc), *libMesh* (http://www.libmesh.github.io), *GSL* (http://www.gnu.org/software/gsl), *MPICH* (http://www.mpich.org), and *Blitz++* (http://www.sourceforge.net/projects/blitz). Finally, all simulations were carried out on a laptop machine—having an Intel Core i7-4600U CPU (@2.10 GHz ×4) and 16GB RAM memory—operating Linux.

RESULTS

In this section we provide some illustrative examples that demonstrate the validity, accuracy, efficiency and robustness of the proposed FE methodology to solve inverse finite deformation analysis problems of biological soft tissues and organs.

Inflation of Cylindrical Tube

The first numerical test we present is the inflation of a thin-wall infinite cylinder. We choose to numerically solve this benchmark problem because it has an ''approximate'' analytic solution, thus making it useful for validation purposes. The cylinder is assumed to be a quasi-incompressible isotropic Neo-Hookean material, having the following stored-energy function: $\bar{W} = \mu/2(J_1 - 3) + \kappa/2(J - 1)^2$, where $\mu = 2.0$ and $\kappa = 10^8$ in units of force per length squared. The inner radius of the cylinder is 20.0 while the thickness is 0.1 units of length, and a uniform pressure load of 0.008 force/length^2 is applied internally. After the application of the load (forward analysis), the deformed cylinder has inner radius 30.024 and thickness 0.0664 units of length. Thus, taking this configuration as an input to the present FE procedure (inverse analysis), the undeformed state of the thin-wall infinite cylinder is evaluated. Due to geometrical symmetry, this problem is analysed using the plane-strain assumption and proper symmetry conditions are applied on the quadrant geometry, while a uniform grid of 400 quadrilateral elements (2 elements in the thickness direction) is constructed.

Five different finite element discretisations for the displacement and the pressure field variables are considered and tested. The accuracy of the algorithm is assessed by comparing the analytical value of the hoop stretch ratio[14]—which is 1.5012— with that computed. The numerical values obtained for the hoop stretch were: (i) 1.50135 when a linear Lagrange basis for the displacements, u, and a constant Monomial basis for the pressure variable, p, were set; (ii) 1.50127 with quadratic Serendipity for u and linear Monomial basis for p; (iii) 1.50134 with quadratic Lagrange for u and linear Monomial basis for p; (iv) 1.50179 in quadratic Serendipity for u and linear Lagrange basis for p; and (v) the exact value was recovered when quadratic and linear Lagrange bases were assumed for the u and p variables respectively.

The inverse FEM achieved very high convergence rates—quadratic convergence in discretisations (iii) to (v)—inherently due to the full-implicit Newton–Raphson scheme implemented. Therefore, in this test and the examples that follow, up to four iterations maximum is all that is required for the solver to converge and reduce the residual by at least nine orders of magnitude. We have investigated the efficiency and accuracy of the proposed algorithm in comparison with the iterative inverse-analysis procedure developed in our group.[9] We observed remarkable reduction of the computational cost due to the fewer iterations needed by the solver to converge. Numerical treatment of material incompressibility is also more robust.

Pinched Torus

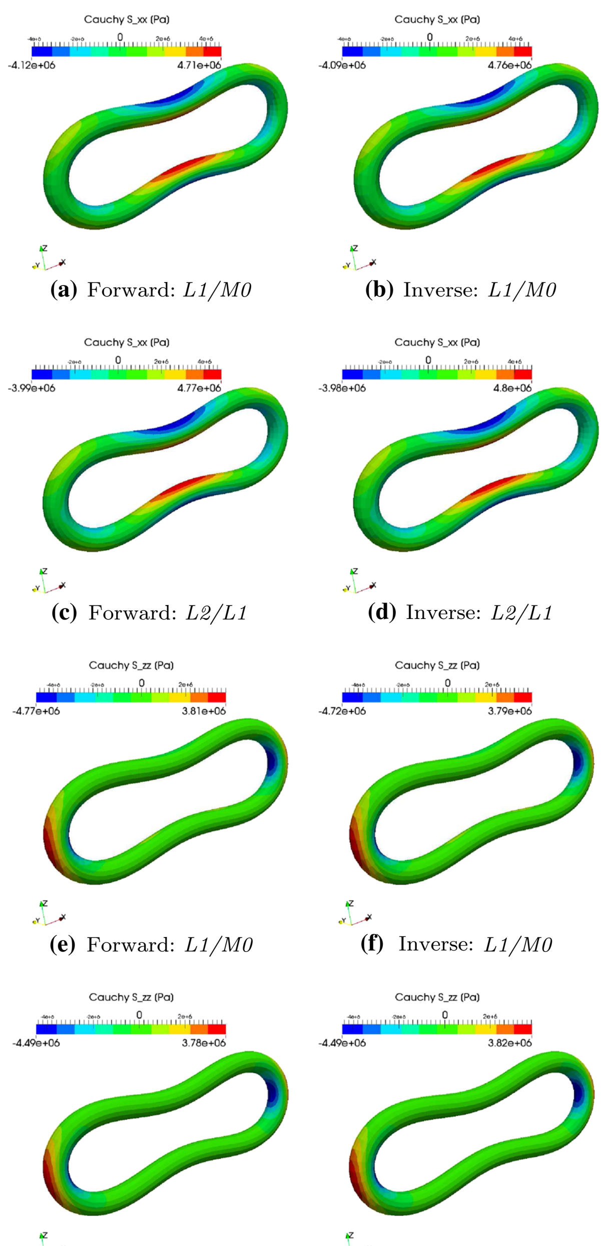
In this test we assume a near-incompressible toroidal solid with dimensions and boundary conditions as in the paper of Chavan *et al.*,[4] where the pinching pressure magnitude is 195 kPa. However, due to symmetry conditions, only one quarter of the toroidal structure was considered and corresponding symmetrical boundary conditions were applied. The quarter torus was coarsely discretised with only 320 triangular-base prism elements using *Gmsh* (http://www.geuz.org/gmsh).The material model used obeyed the stored energy function given by: $\bar{W} = c_1(J_1 - 3) - \kappa \log(J) + \kappa/2\,(J^2 - 1)$, where the material parameters used are $c_1 = 6$ MPa and $\kappa = 30$ GPa.

In this example, we initially carried out forward analysis[37] simulations on the original (undeformed) torus, where different basis function cases were considered: (a) first-order Lagrangian nodal basis for displacements and constant Monomial modal basis for the hydrostatic pressure (abbreviated: *L1/M0*), and (b) second- and first-order Lagrangian basis for the displacement and pressure variables respectively (abbreviated: *L2/L1*). Then, the FE mesh of the toroidal geometry was updated—based on the obtained numerical solution of the nodal displacements—and was subsequently utilised by the inverse analysis tool. Hence, we attempted to validate the inverse FE solver by recovering the original shape of the pinched torus and predicting the initial stresses.

The performance of the algorithm is assessed by calculating the difference of the space vector of all nodes in the undeformed mesh with respect to the nodes of the ''updated'' mesh resulting from the inverse analysis. In the first case (*L1/M0*) the median space vector difference is approximately 10 mm while for the second analysis case (*L2/L1*) it is 7 mm. The level of accuracy of the inverse algorithm is evident by comparing the above findings with the actual size of the torus (10 m outer radius). This verifies the improved accuracy achieved when using 2nd-order elements and assuming a continuous p-field. We also compared the numerically predicted maximum principal stretch ratio, $\lambda_{\max}$, in both forward and inverse analyses. In the conventional forward cases *L1/M0* and *L2/L1* $\lambda_{\max}$ ranged: 1.022–1.281 and 1.018–1.287

respectively, while excellent correspondence is achieved by the corresponding inverse simulations: 1.023–1.287 and 1.018–1.289 respectively.

The σ_{xx} and σ_{zz} components of the Cauchy stress tensor—obtained from the forward analysis— plotted on the final deformed configuration are shown in Figs. 1a–1g. Also, in Figs. 1b–1h, contour plots of the same Cauchy stress components are presented that correspond to the predictions of the initial stress distribution based on the proposed inverse design method. It can be observed from these results that volumetric locking is effectively avoided when utilising a mixed FE formulation for the stress analysis (forward and inverse) of quasi-incompressible solids, which does not hold when a conventional displacement based FE formulation is used (not shown here).However, we expect the numerical predictions for the inverse analysis simulations to be slightly degraded, since the FE grid used was obtained directly from the forward analysis; thus, the current (deformed) FE mesh had inherited the element distortions from the results obtained from the previous (forward) analysis.



(a) Forward: *L1/M0* **(b)** Inverse: *L1/M0*

(c) Forward: *L2/L1* **(d)** Inverse: *L2/L1*

(e) Forward: *L1/M0* **(f)** Inverse: *L1/M0*

(g) Forward: *L2/L1* **(h)** Inverse: *L2/L1*

FIGURE 1. Comparison of the predicted Cauchy stress distribution on the final deformed shape of the pinched torus using standard forward analysis (left column) with the inverse design analysis computed pre-stress field (right column).

Aortic Arch

The third numerical example we investigate is the recovery of the zero-pressure state of the ascending thoracic aorta. The geometry of an aortic arch of a healthy volunteer is reconstructed based on CT-scans (see black wireframe in Fig. 2) and discretised with 15-node second-order prismatic elements.[24] In total, the finite element grid consisted of 12,455 prism elements and 93,671 nodes. The outlets of the innominate artery, the left carotid and left subclavian arteries are not allowed to translate perpendicular to the plane defined by these outlets. The inlet of the ascending aorta and the outlet of the descending aorta are fixed to avoid rigid body motion. In addition, prescribed pressure is applied to the lumen area and the external surface of the aorta is assumed traction-free. The pressure mag-

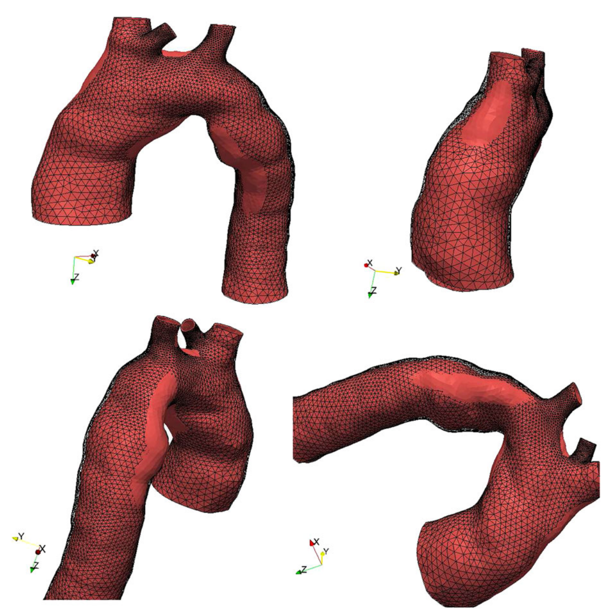


FIGURE 2. Comparison of the *in vivo* geometry of the thoracic aorta (black wireframe) and the corresponding zero-pressure configuration (red surface).

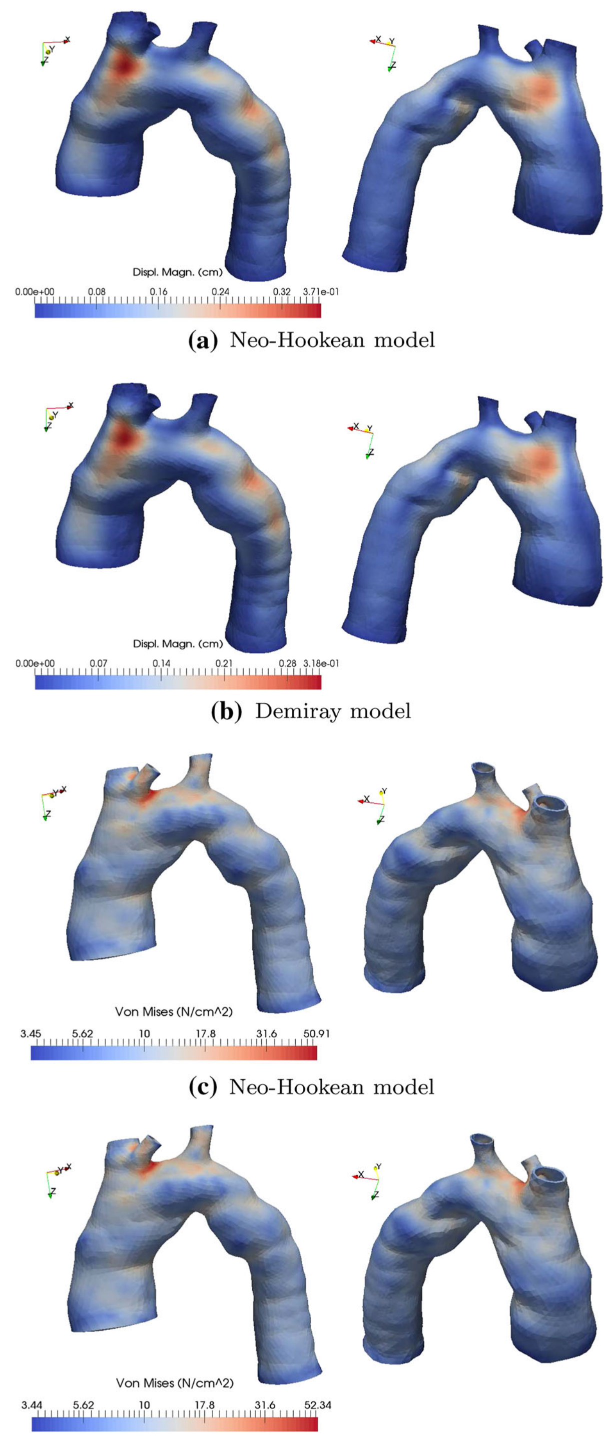
nitude is taken to be equal to the mean aortic pressure, which for resting humans can be better approximated by the geometric mean of the systolic and diastolic aortic pressure.[5] Thus, a 100 mmHg uniform internal pressure loading is considered and applied in ten equispaced increments.

According to the experiments of Mosora *et al.*,[27] the Young's modulus of the ascending thoracic aorta varied from 2 to 6 MPa. In addition, Xie *et al.*[41] performed bending experiments and observed the Young's modulus of the inner layer of the aorta (i.e., intima and media) to be three times larger than of the outer layer (adventitia).In the present study, a homogeneous, isotropic constitutive law is assumed for the aortic wall, based on the following expressions: $\bar{W} = \mu/2\,(J_1 - 3) - \mu \log J + K/2\,(J-1)^2$ and $\bar{W} = C_1/C_2\,[e^{C_2(J_1-3)/2} - 1] + K/2\,(J-1)^2$. The material parameters μ and K for the modified Neo-Hookean model are chosen to match the small-strain shear and bulk modulii, i.e., 1 MPa and 50 GPa respectively, while for the Demiray model the parameters C_1, C_2 are taken from the literature.[36]

In Fig. 3a, contours of the magnitude of the displacement vector are shown based on the inverse deformation analysis of the thoracic aorta when modelled as Neo-Hookean, while comparison of the aortic wall shape of the originally acquired deformed configuration and the predicted unloaded state is illustrated in Fig. 2. As seen from this figure, pronounced deformation occurs on the lateral (both in the posterior and anterior) walls of the aortic arch, which is due to the local reduction in the thickness of the aortic wall. However, both models predicted relatively similar zero-pressure configurations of the aortic arch geometry, with the Demiray model (see Fig. 3b) giving a relatively stiffer arterial wall. Additionally, the von Mises stress distribution is depicted in Figs. 3c and 3d, showing both models produced pronounced prestressing at the origin of the brachiocephalic and the left common carotid artery.

Abdominal Aortic Aneurysm

Here we consider a patient-specific AAA that contains a thin layer of intraluminal thrombus (ILT). The internal radius and the thickness of the common iliac arteries are 6.7 mm and 1.8 mm respectively. The thrombotic layer is about 1 mm thick, the internal radius of the abdominal aorta above the sac 11 mm and the thickness of the ILT is approximately 0.6 mm. A 2nd-order polynomial constitutive law with respect to the first and second invariant of the deformation tensor is adopted for the biomechanical description of the vessel wall and the thrombus[26] respectively, where the material coefficients are taken from the experimental works of Raghavan and Vorp[31] and Wang



(a) Neo-Hookean model

(b) Demiray model

(c) Neo-Hookean model

(d) Demiray model

FIGURE 3. Contour plot of the displacement magnitude (in cm) and the von Mises stress (in N/cm^2) of the zero-pressure thoracic aorta configuration.

et al.[39] However, it is important to note here that a major limitation in the present analysis is the assumption of uniform wall thickness, which is due to

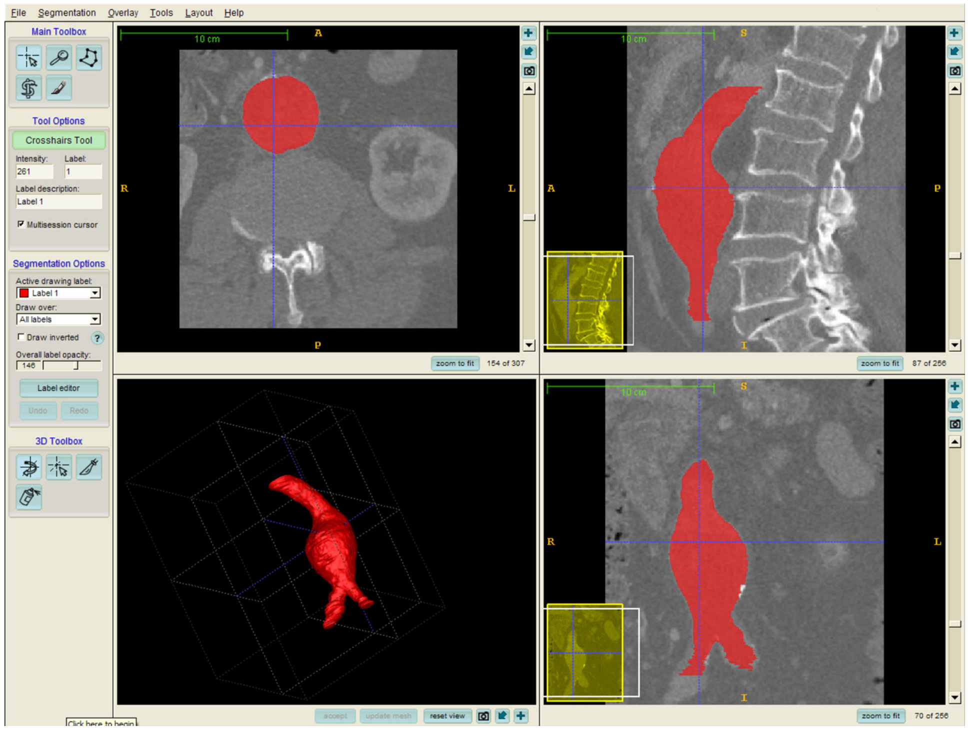


FIGURE 4. Segmented CT images of the patient-specific AAA geometry visualised using *itk-SNAP*.

the limitations in the ability of existing imaging tools to accurately measure this.

The geometry of the AAA was reconstructed using CT scans based on which the aortic wall and the thrombotic region are segmented. Figure 4 illustrates a screenshot of the segmentation outcome of the CT image data, which were processed using *itk-SNAP* (http://www.itksnap.org). Subsequently, a three-dimensional finite element grid is constructed by surface extrusion having 15,748 prism elements and 39,690 nodes.The inlet and outlet rings are clamped, the external surface of the wall is traction-free and an internal pressure load is applied at the lumen surface, similarly to the previous example.

In Fig. 5, direct comparison of the original CT acquired AAA geometry with the corresponding inverse analysis predicted pressure-free geometry is illustrated, when the AAA–ILT geometry is considered in the inverse analysis (see Fig. 5a) and when the ILT is neglected in the FE simulation (see Fig. 5b).

The von Mises stress of the AAA is depicted in Figs. 6a and 6b, which corresponds to the pre-stressing of the structure due to the *in vivo* conditions. Von Mises stress distribution ranged from 0.35 N cm^{-2} to about 22.4 N cm^{-2}. It can be observed from these figures that the ILT region bears a very low pre-stress distribution as opposed to the aortic wall seen in Fig. 6a. Qualitatively similar results were published by Gee *et al.*[11] although the exact value range of their predictions of

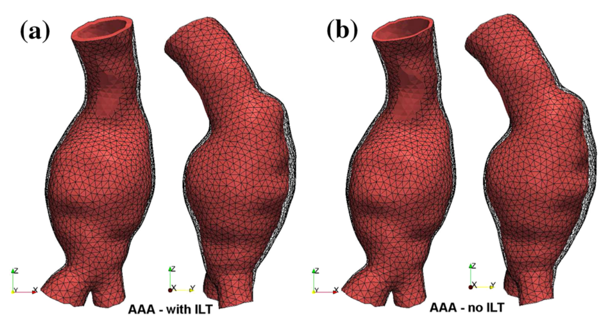
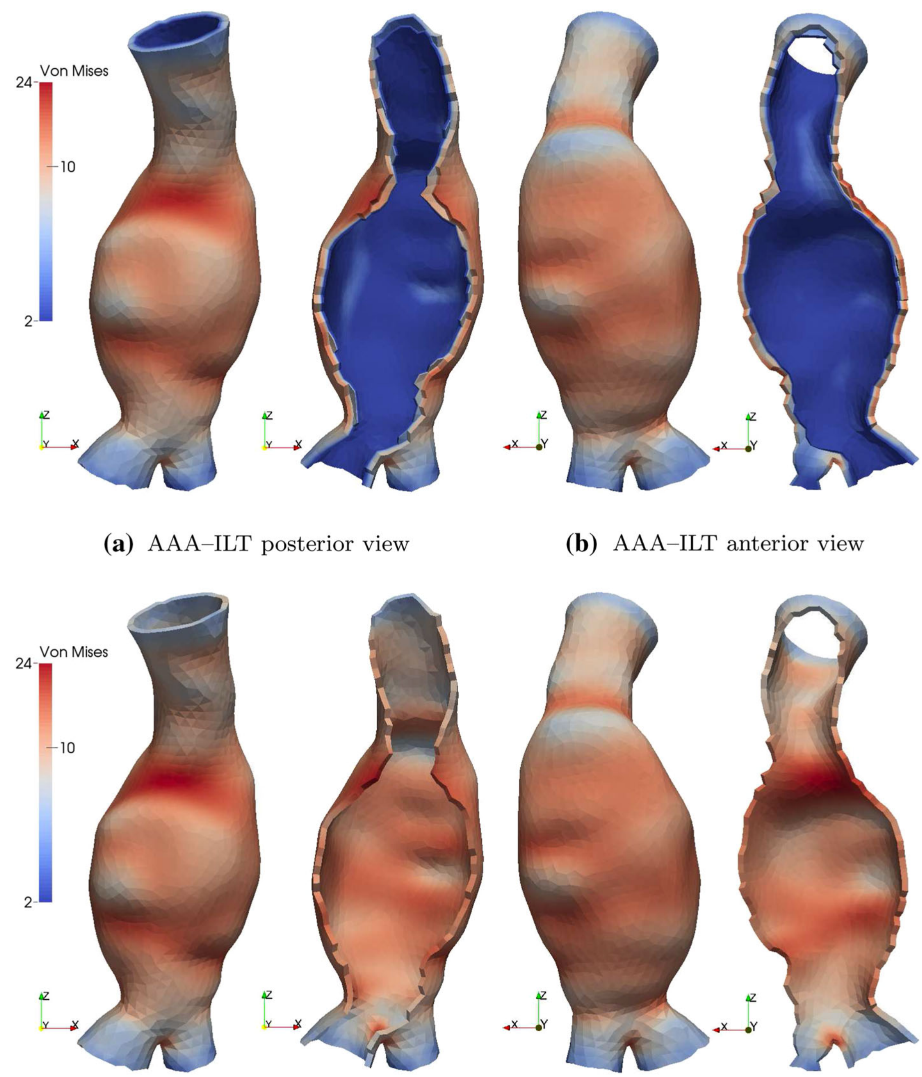


FIGURE 5. Comparison of the *in vivo* AAA geometry (black wireframe) and the corresponding zero-pressure configuration (red surface).

the von Mises stress is not reported. In addition, in Figs. 6c and 6d, von Mises distribution is also shown in the special case when the ILT region is virtually removed from the analysed domain. The von Mises stresses ranged from 2.59 N/cm^2 to approximately 24.4 N/cm^2. Quantitatively similar results were reported in Ref. 21 for the inverse simulation of an AAA without the presence of ILT. However, existing experimental studies have demonstrated that arterial wall tissue exhibits orthotropic mechanical response,[31] mainly attributed to the elastin and collagen fibres deposition pattern. In order to further investigate the influence of directional isotropy on the initial stress predictions, the arterial wall is reinforced by considering two sets of fibres, i.e., $\hat{\mathbf{n}}_0$ and $\hat{\mathbf{n}}_1$.Therefore, anisotropic



(a) AAA–ILT posterior view **(b)** AAA–ILT anterior view

(c) AAA posterior view **(d)** AAA anterior view

FIGURE 6. Initial von Mises stress distribution (colour scales in N/cm^2) of the AAA geometry when (a, b) the ILT layer is present and (c, d) the ILT is neglected.

contribution is modelled by adding two extra terms in $\bar{W}$ as a function of the modified pseudo-invariants J_4 and J_6.[16,35] Fibres are chosen to align $\pm 33^\circ$ to the local horizontal axis in the tangent plane, with the latter being determined during image processing. Interestingly, inverse analysis simulation results (not presented here) of the anisotropic model AAA—with and without the presence of ILT—gave rise to the maximum von Mises pre-stress level at about only 1%. Hence, anisotropy made only a minor contribution to the inverse analysis prediction of the zero-pressure AAA shape, which is in agreement with the results of Riveros *et al.*

Finally, in view of Figs. 5 and 6, very important observations can be drawn with respect to the predominance of the ILT in the stress analysis of AAA geometries and the pre-stressing estimation. In Figs. 5a and 5b, it is clearly demonstrated that the presence of the ILT layer can give different aortic-wall reference configuration predictions, hence, significantly affecting subsequent studies in rupture risk assessment and haemodynamic analysis, as also reported by various other investigators.[2,26,40]

Breast Deformation

The female breast can undergo large deformations during routine screening, diagnostic imaging, therapy planning or surgical procedures. For instance, MRI is a commonly used imaging modality for breast cancer diagnosis with the individual positioned prone; whereas in CT scanning for radiotherapy planning, the patient is usually supine. As a natural consequence of this, the

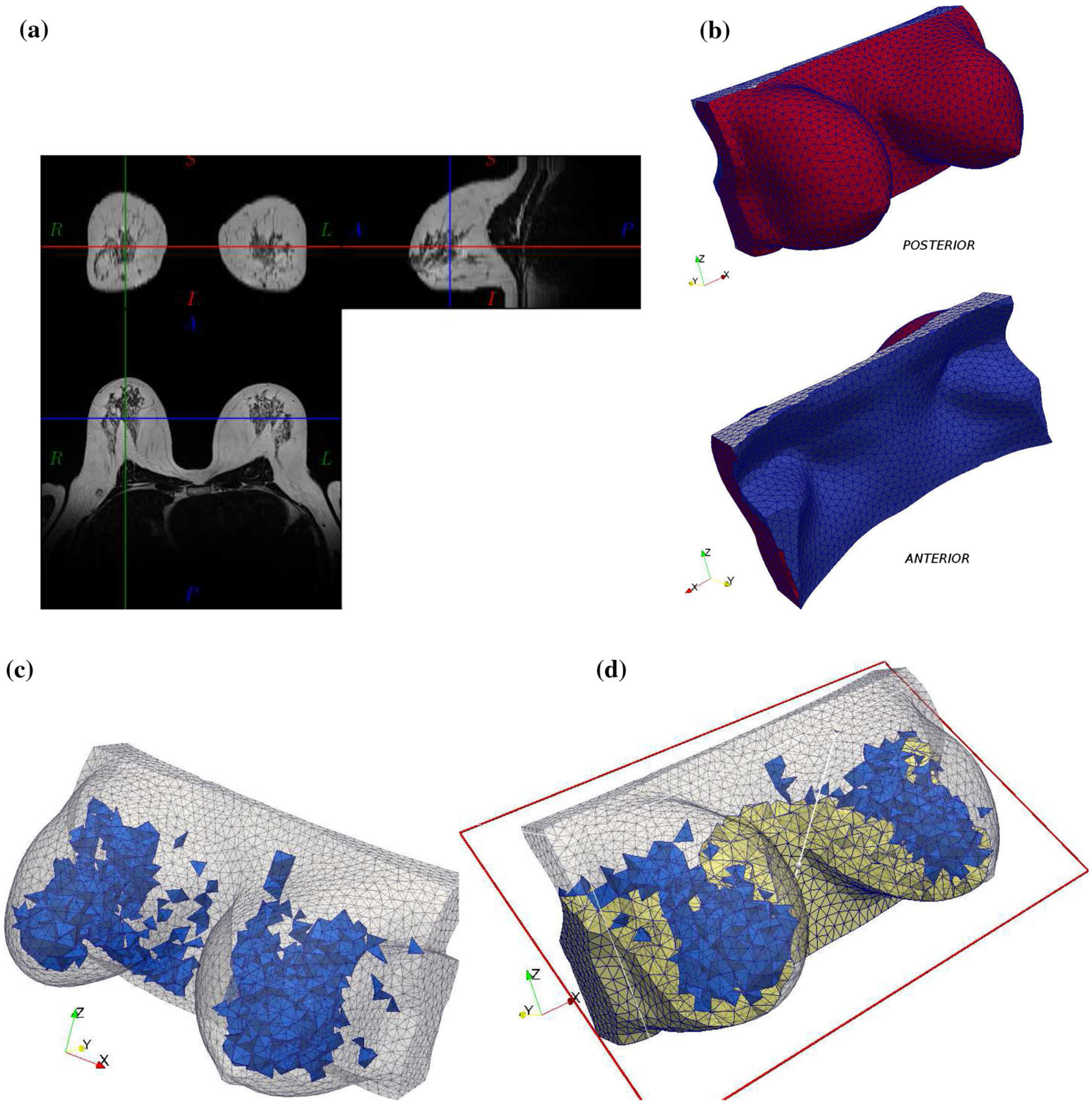


FIGURE 7. (a) Orthogonal view of the MR-acquired image of the patient-specific breast. (b) Discretised breast geometry in prone configuration and (c) internal mesh structure of fibroglandular tissues (blue tetrahedral elements) and a (d) clipped mesh of adipose tissues (yellow tetrahedral elements).

breast deforms due to the effect of gravity. Therefore in this example, we attempt to recover the zero-gravity configuration of the breast of an adult female.

In the present inverse analysis simulation, the breast/chest geometry of a healthy volunteer in prone position is considered. MR images were acquired (see Fig. 7a) and segmented into the various tissue groups (i.e., adipose and fibroglandular tissues, muscle and skin). However, further details on the adopted segmentation procedure can be found in the recent paper of Eiben and his colleagues,[9] while *Gmsh* is utilised for the mesh generation comprising of 7067 second-order tetrahedral elements.

To date, various constitutive models have been adopted to describe the biomechanics of breast tissues. These include, isotropic and transversely isotropic models, based either on a simple linear elastic law or on more complex hyperelastic constitutive descriptions.[8,15] However, in the present example we assumed a simplified isotropic Neo-Hookean model for the biomechanical description of breast tissues, based on the relation: $\bar{W} = c_1\,(J_1 - 3) + 1/D\,(J - 1)^2$. Adipose (fat) and fibroglandular tissues are considered separately, where the material parameter c_1 is 0.08 and 0.12 kPa respectively, and D is set to 10 MPa $^{-1}$ for both tissue groups. In the present inverse analysis we also accounted for the dermis/epidermis contribution to the breast deformation. The skin is modelled as a membrane (red triangularized surface in Fig. 7b) with uniform thickness 1.5 mm, obeying a polynomial hyperelastic material law and material properties as in del Palomar *et al.*[8] In this

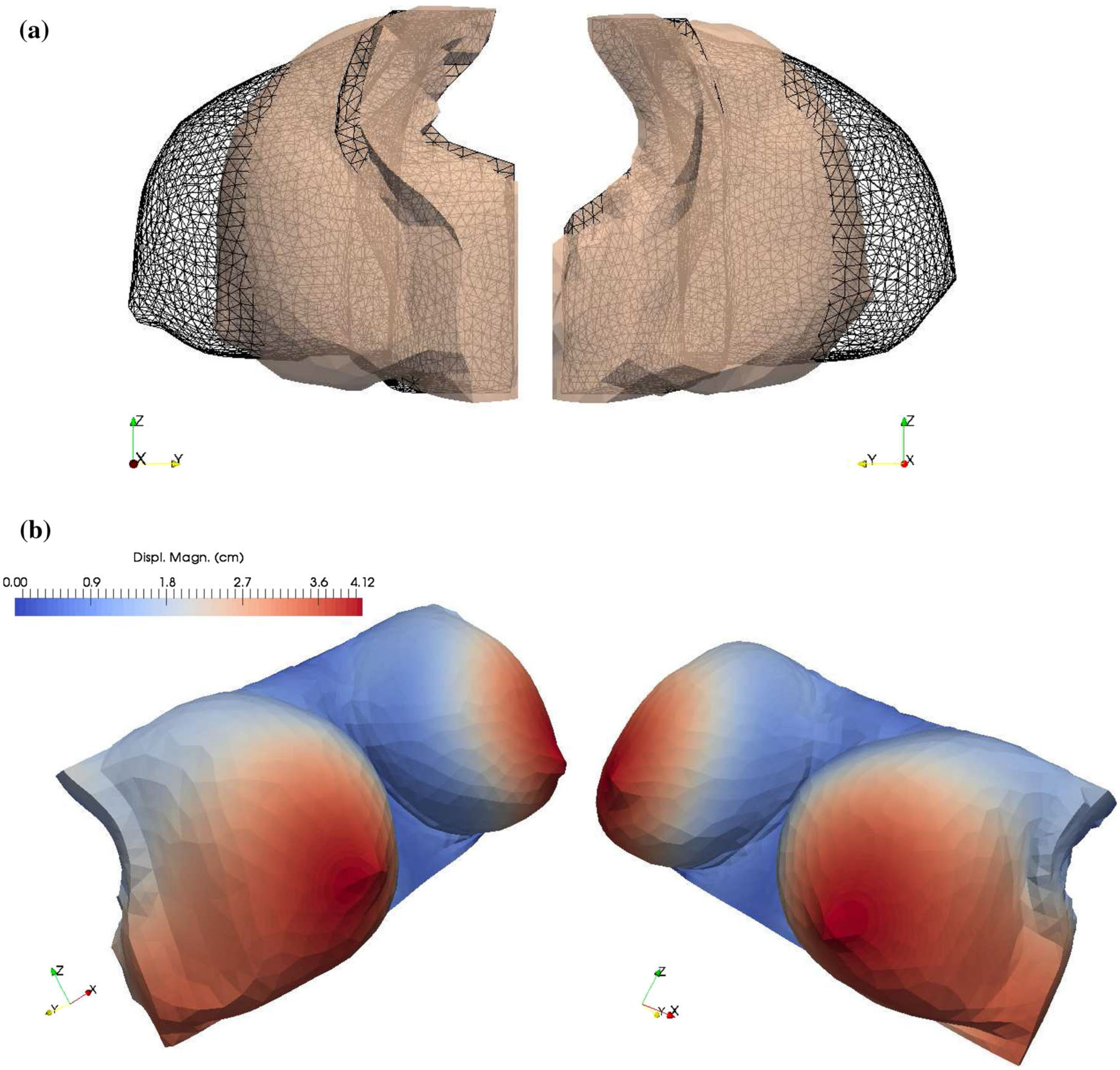


FIG. 8. (a) Comparison of the loaded (black wireframe) and gravity-free (transparent surface) breast geometry and (b) contour plot of the displacement magnitude (in cm) computed from the inverse analysis overlaid on the undeformed breast configuration.

regard, we implemented the inverse deformation analysis for the membrane elements of Lu *et al.*[22]

The boundary conditions associated with this problem were fixed displacements on the fascial boundary (blue surface in Fig. 7b), fixed displacement in the axial direction (i.e., $u_Z = 0$) on the superior and inferior horizontal planes (grey surface in Fig. 7b), and the skin was traction-free. In addition, the gravity load vector was set to the posterior to anterior direction.

In Fig. 8a, the initial breast shape in the prone setting (black wireframe) is superimposed on the predicted inverse-design reference state. Also in Fig. 8b, the magnitude of the numerically evaluated displacement field in centimetres is also illustrated. However, final examination of the results obtained with the inverse method is carried out. Given the reference (zero-gravity) configuration of the breast geometry, forward analysis is performed in order to predict the breast shape in the prone setting. After evaluating the differences in nodal positions between the original MR-based geometry and the newly predicted prone breast shape, a maximum error of 0.03 mm is computed which verifies the excellent consistency of the algorithm.

Evidently, the above predicted zero-gravity configuration of the patient-specific breast geometry can be used as a starting point for subsequent forward analysis simulations in order to predict the breast shape in standing position or in supine, which is usually the setting in the operating theatre.

DISCUSSION

In this article, we propose a two-field Eulerian formulation based on the finite element method for inverse problem analysis. The scope of this analysis approach is to recover the stress-free or unloaded state of an *in vivo* biological organ or soft tissue given its current deformed configuration, boundary conditions and the biomechanical properties of the tissues. The three-dimensional inverse problem posed here is (as in the forward counterpart) static-determinate and equilibrium can be achieved given a predefined set of loading and material parameters. As a result, numerical determination of the initial configuration of the biological solid structure is directly influenced by the decision of the patient-specific parameters of the model.However, this approach is founded on the concept of re-parametrisation of the balance equation in the inverse setting,[13] where hydrostatic pressure is introduced here as an extra unknown term. Alternatively it can be assumed to be continuous across all finite elements and can incorporate more elaborate volumetric constitutive expressions, as opposed to the simplifications of Govindjee and Mihalic[14] and Lu *et al.*[20]

The present method encounters no difficulties reaching a stable solution of the reference configuration as opposed to existing techniques modelling arterial wall inverse biomechanics using shell or membrane elements.[11,19,22,43] The method is fully implicit thus achieving quadratic convergence, while simulations require very few load steps to reach final solution compared to explicit solvers,[17] and it can run quickly since no mesh updating is required during the analysis.[2,11] Additionally, the present algorithm ensures convergence using a combination of residual and energy criterion checks. In contrast to this, existing backward incremental methods or pull-back iterative inverse approaches[7,9,33,34] are limited with respect to nodal spatial-location convergence checks. As a result such criteria in load-control nonlinear problems, such as the pressurisation of the aorta, cannot guarantee satisfaction of the force-balance numerically.

The proposed inverse design methodology can be readily incorporated into a standard forward analysis Updated Lagrangian FE code[37] by carefully changing the routine that evaluates of the tangent stiffness matrix and the internal force contribution in the *rhs* vector. The validity and accuracy of the numerical methodology was successfully tested on benchmark problems and utilised in three bioengineering applications. We predicted numerically the pressure-free configuration and the initial stress distribution of an aortic arch and a patient-specific AAA, and recovered the gravity-free state of a female breast geometry. The present numerical scheme has been tested in recovering the unloaded configuration of various subject- and patient-specific problems (not presented here).It has been observed to perform robustly with very reasonable computational time cost. The overall computational time spent to realise the patient-specific simulations ("Aortic arch" to "Breast deformation" sections) varied between 8 and 12 min, depending on the total number of degrees-of-freedom in the system, which is very good. However, computational cost can be further reduced if code becomes fully optimised and simulations are carried out in a computer cluster, especially when high-dense FE discretisations are required. Additionally, this framework can be also applied to a variety of inverse analysis bioengineering problems, where soft tissues or organs are under external loads, e.g., cerebral aneurysms, urinary bladder, as well as image registration for lung and brain mapping, and in surgical modelling for pre-operative planning.

APPENDIX: MATHEMATICAL DERIVATIONS

The second Piola-Kirchhoff stress tensor for an elastic solid that is described by the modified stored-energy potential $\bar{W} + Q$[37] is given by

$$\boldsymbol{S} = \boldsymbol{S} - \frac{2(\bar{p} - p)}{P(\bar{p})}\frac{\partial \bar{p}}{\partial \mathbf{C}} + \frac{(\bar{p} - p)^2}{P^2(\bar{p})}\frac{\partial P(\bar{p})}{\partial \mathbf{C}}, \tag{11}$$

where $\boldsymbol{S} = 2\,\partial \bar{W}/\partial \mathbf{C}$, $\bar{p} = -\partial \bar{W}/\partial J$, and $P(\bar{p}) = \partial \bar{p}/\partial J$. Applying the push-forward transformation $(J\boldsymbol{\sigma} = \mathbf{F}\cdot\mathbf{S}\cdot\mathbf{F}^T)$ on the above formula and considering that: $\mathbf{F}\cdot[\partial\bullet/\partial\mathbf{C}]\cdot\mathbf{F}^T = \mathbf{B}\cdot[\partial\bullet/\partial\mathbf{B}]$, we obtain the Cauchy stress tensor for the forward problem

$$\boldsymbol{\sigma} = \boldsymbol{\sigma} - \frac{2(\bar{p} - p)}{JP(\bar{p})}\mathbf{B}\cdot\frac{\partial \bar{p}}{\partial \mathbf{B}} + \frac{(\bar{p} - p)^2}{JP^2(\bar{p})}\mathbf{B}\cdot\frac{\partial P(\bar{p})}{\partial \mathbf{B}}. \tag{12}$$

Re-parametrisation of the Cauchy stresses expression to an inverse setting,[13] i.e., replacing $J = \jmath^{-1}$ and $\mathbf{B} = \mathbf{c}^{-1}$, gives

$$\begin{aligned}\boldsymbol{\sigma} &= \boldsymbol{\sigma} - \frac{2\jmath(\bar{p} - p)}{P(\bar{p})}\mathbf{c}^{-1}\cdot\frac{\partial \bar{p}}{\partial \mathbf{c}^{-1}} + \frac{\jmath(\bar{p} - p)^2}{P^2(\bar{p})}\mathbf{c}^{-1}\cdot\frac{\partial P(\bar{p})}{\partial \mathbf{c}^{-1}} \\ &= \boldsymbol{\sigma} - \frac{2\jmath(\bar{p} - p)}{P(\bar{p})}\mathbf{c}^{-1}\cdot\frac{\partial \bar{p}}{\partial J}\frac{\partial \jmath^{-1}}{\partial \mathbf{c}^{-1}} \\ &\quad + \frac{\jmath(\bar{p} - p)^2}{P^2(\bar{p})}\mathbf{c}^{-1}\cdot\frac{\partial P(\bar{p})}{\partial J}\frac{\partial \jmath^{-1}}{\partial \mathbf{c}^{-1}},\end{aligned} \tag{13}$$

where after some algebra it can be deduced that

$$\boldsymbol{\sigma} = \boldsymbol{\sigma} - \frac{\bar{p} - p}{P(\bar{p})}\frac{\partial \bar{p}}{\partial J}\mathbf{I} + \frac{(\bar{p} - p)^2}{2P^2(\bar{p})}\frac{\partial P(\bar{p})}{\partial J}\mathbf{I}. \tag{14}$$

In the forward setting, the Cauchy stress tensor is defined by Eq. (1), where the potential stored-energy is a function of the modified strain invariants, i.e., $\bar{W}(J_1, J_2, J)$. However, in the particular case of a transversely isotropic hyperelastic solid having one family of fibres—described by a single direction unit vector ($\hat{\mathbf{N}}_0$) in the reference state—the potential stored-energy can be extended to include the modified pseudo-invariants:[16] $J_4 = J^{-2/3}\,\hat{\mathbf{N}}_0 \cdot \mathbf{C} \cdot \hat{\mathbf{N}}_0$ and $J_5 = J^{-4/3}\,\hat{\mathbf{N}}_0 \cdot \mathbf{C}^2 \cdot \hat{\mathbf{N}}_0$. Apparently more pseudo-invariants can be incorporated in $\bar{W}$ if more preferred directions of isotropy are considered in the constitutive model.

Using chain rule differentiation in Eq. (1), substituting the tensor derivatives of the modified invariants into Eq. (1) and making use of the Cayley–Hamilton theorem[23] for $\mathbf{B}$, i.e., $\mathbf{B}^2 = I_1\,\mathbf{B} - I_2\,\mathbf{I} + J^2\,\mathbf{B}^{-1}$, produces

$$
\begin{aligned}
\boldsymbol{\sigma} = {} & \frac{\partial \bar{W}}{\partial J_1}\left(2J^{-5/3}\,\mathbf{B} - 2J_1/3J\mathbf{I}\right) \\
& + \frac{\partial \bar{W}}{\partial J_2}\left(2J_2/3J\mathbf{I} - 2J^{-1/3}\,\mathbf{B}^{-1}\right) + \frac{\partial \bar{W}}{\partial J}\,\mathbf{I} \\
& + \frac{\partial \bar{W}}{\partial J_4}\left(2J^{-5/3}\,\mathbf{A}_0\right) + \frac{\partial \bar{W}}{\partial J_5} \\
& \times \left(2J^{-7/3}\,\mathbf{A}_0 \cdot \mathbf{B} + 2J^{-7/3}\,\mathbf{B} \cdot \mathbf{A}_0\right).
\end{aligned}
\tag{15}
$$

where $\mathbf{A}_0 = \mathbf{F} \cdot (\hat{\mathbf{N}}_0 \otimes \hat{\mathbf{N}}_0) \cdot \mathbf{F}^T$. However, it is straightforward to extend this constitutive equation for multi-fibrous anisotropic hyperelastic materials by incorporating more pseudo-invariants in the isochoric part of the potential function.

Re-parametrisation of the above Cauchy stresses expression to an inverse setting can give the analytic expression

$$
\begin{aligned}
\boldsymbol{\sigma} = {} & \frac{\partial \bar{W}}{\partial J_1}\left(2\jmath^{5/3}\,\mathbf{c}^{-1} - 2/3\jmath_2\,\jmath\,\mathbf{I}\right) \\
& + \frac{\partial \bar{W}}{\partial J_2}\left(2/3\jmath_1\,\jmath\mathbf{I} - 2\jmath^{1/3}\,\mathbf{c}\right) + \frac{\partial \bar{W}}{\partial J}\,\mathbf{I} \\
& + \frac{\partial \bar{W}}{\partial J_4}\left(2\,\jmath_4^{-1}\,\jmath\mathbf{a}_0\right) + \frac{\partial \bar{W}}{\partial J_5} \\
& \times \left(2\,\jmath_4^{-1}\,\jmath^{5/3}\,\mathbf{a}_0 \cdot \mathbf{c}^{-1} + 2\,\jmath_4^{-1}\,\jmath^{5/3}\,\mathbf{c}^{-1} \cdot \mathbf{a}_0\right),
\end{aligned}
\tag{16}
$$

where $J_1 = \jmath_2$, $J_2 = \jmath_1$, $J_4 = \jmath_4^{-1}$, $J_5 = \jmath_5\,\jmath_4^{-1}\,\jmath^2$, $J = \jmath{-}1$ ($\jmath = \det \mathbf{f}$), and $\jmath_1 = \imath_1\,\jmath^{-2/3}$, $\jmath_2 = \imath_2\,\jmath^{-4/3}$, $\jmath_4 = \imath_4\,\jmath^{-2/3}$, $\jmath_5 = \imath_5\,\jmath^{-4/3}$, while $\mathbf{a}_0 = \hat{\mathbf{n}}_0 \otimes \hat{\mathbf{n}}_0$. The first and second invariants of the inverse deformation tensor $\mathbf{c}$ are denoted with $\imath_1$ and $\imath_2$ respectively, while $\imath_4$, $\imath_5$ are the corresponding Eulerian pseudo-invariants of the fibre.[20] Also, the fibre tangent unit vector in the deformed state, $\hat{\mathbf{n}}_0$, is related to the corresponding one in the reference frame, $\hat{\mathbf{N}}_0$, through the relationship:

$$
\begin{aligned}
& \mathbf{F} \cdot \hat{\mathbf{N}}_0 = I_4^{1/2}\hat{\mathbf{n}}_0 \Leftrightarrow \hat{\mathbf{N}}_0 = I_4^{1/2}\mathbf{F}^{-1} \cdot \hat{\mathbf{n}}_0 \\
& \quad \Leftrightarrow \hat{\mathbf{N}}_0 = \imath_4^{-1/2}\mathbf{f} \cdot \hat{\mathbf{n}}_0 \Leftrightarrow \hat{\mathbf{N}}_0 = \jmath_4^{-1/2}\,\jmath^{-1/3}\mathbf{f} \cdot \hat{\mathbf{n}}_0,
\end{aligned}
$$

In order to arrive at Eq. (2), one has to substitute the partial derivatives of the stored-energy function with respect to the inverse invariants, i.e., $\partial\bar{W}/\partial J_1 = \partial\bar{w}/\partial\jmath_2$, $\partial\bar{W}/\partial J_2 = \partial\bar{w}/\partial\jmath_1$, $\partial\bar{W}/\partial J = -\jmath^2\,\partial\bar{w}/\partial\jmath$, $\partial\bar{W}/\partial J_4 = -\jmath_4^2\,\partial\bar{w}/\partial\jmath_4 - \jmath_4\,\jmath_5\,\partial\bar{w}/\partial\jmath_5$ and $\partial\bar{W}/\partial J_5 = \jmath_4\,\jmath^2\,\partial\bar{w}/\partial\jmath_5$, in Eq. (16).

Finally, taking the above equation and after some tedious algebra, the following analytic expression for the material tangent fourth-order tensor of Eq. (10b) can be deduced:

$$
\begin{aligned}
\bar{\mathsf{C}}_{\mathrm{UU}} = {} & \left[\frac{\partial^2 \bar{W}}{\partial J_1^2}\left(4\,\jmath^{5/3}\,\mathbf{c}^{-1} - 4/3\jmath_2\,\jmath\,\mathbf{I}\right)\right. \\
& \left. + \frac{\partial^2 \bar{W}}{\partial J_2 \partial J_1}\left(4/3\jmath_1\,\jmath\,\mathbf{I} - 4\jmath^{1/3}\,\mathbf{c}\right) + \frac{\partial^2 \bar{W}}{\partial J \partial J_1}\,2\,\mathbf{I}\right]\frac{\partial \jmath_2}{\partial \mathbf{c}} \\
& + \left[\frac{\partial^2 \bar{W}}{\partial J_1 \partial J_2}\left(4\jmath^{5/3}\,\mathbf{c}^{-1} - 4/3\jmath_2\,\jmath\,\mathbf{I}\right)\right. \\
& \left. + \frac{\partial^2 \bar{W}}{\partial J_2^2}\left(4/3\jmath_1\,\jmath\,\mathbf{I} - 4\jmath^{1/3}\,\mathbf{c}\right) + \frac{\partial^2 \bar{W}}{\partial J \partial J_2}\,2\,\mathbf{I}\right]\frac{\partial \jmath_1}{\partial \mathbf{c}} \\
& - \frac{\partial^2 \bar{W}}{\partial J_1 \partial J}\left(2\jmath^{2/3}\,\mathbf{c}^{-1} - 2/3\jmath_2\,\mathbf{I}\right)\mathbf{c}^{-1} \\
& - \frac{\partial^2 \bar{W}}{\partial J_2 \partial J}\left(2/3\jmath_1\,\mathbf{I} - 2\jmath^{-2/3}\,\mathbf{c}\right)\mathbf{c}^{-1} \\
& + \frac{\partial \bar{W}}{\partial J_1}\left(20/3\jmath^{2/3}\,\mathbf{c}^{-1}\,\frac{\partial \jmath}{\partial \mathbf{c}} + 4\jmath^{5/3}\,\mathsf{I}_{c^{-1}}\right. \\
& \left. - 4/3\jmath\,\mathbf{I}\,\frac{\partial \jmath_2}{\partial \mathbf{c}} - 4/3\jmath_2\,\mathbf{I}\,\frac{\partial \jmath}{\partial \mathbf{c}}\right) \\
& + \frac{\partial \bar{W}}{\partial J_2}\left(4/3\jmath\,\mathbf{I}\,\frac{\partial \jmath_1}{\partial \mathbf{c}} + 4/3\jmath_1\,\mathbf{I}\,\frac{\partial \jmath}{\partial \mathbf{c}}\right. \\
& \left. - 4/3\jmath^{-2/3}\,\mathbf{c}\,\frac{\partial \jmath}{\partial \mathbf{c}} - 4\jmath^{1/3}\,\mathsf{I}\right) \\
& - \frac{\partial^2 \bar{W}}{\partial J_4 \partial J_4}\left(4\,\jmath/\jmath_4^3\mathbf{a}_0\,\frac{\partial \jmath_4}{\partial \mathbf{c}}\right) \\
& - \frac{\partial^2 \bar{W}}{\partial J_4 \partial J_5}\left(4\,\jmath/(\jmath_4\,\jmath_5^2)\mathbf{a}_0\,\frac{\partial \jmath_5}{\partial \mathbf{c}}\right) \\
& + \frac{\partial \bar{W}}{\partial J_4}\left(4/\jmath_4\mathbf{a}_0\,\frac{\partial \jmath}{\partial \mathbf{c}} - 4\,\jmath/\jmath_4^2\mathbf{a}_0\,\frac{\partial \jmath_4}{\partial \mathbf{c}}\right) \\
& + \frac{\partial \bar{W}}{\partial J_5}\left(10/3\jmath^{5/3}/\jmath_4\mathsf{N}_{0c^{-1}}\,\mathbf{c}^{-1}\right. \\
& \left. - 4\,\jmath^{5/3}/\jmath_4^2\mathsf{N}_{0c^{-1}}\,\frac{\partial \jmath_4}{\partial \mathbf{c}} + 4\,\jmath^{5/3}/\jmath_4\mathsf{N}_{0c^{-1}}\right) \\
& - \frac{\partial^2 \bar{W}}{\partial J_5 \partial J_4}\left(4\,\jmath^{5/3}/\jmath_4^3\mathsf{N}_{0c^{-1}}\,\frac{\partial \jmath_4}{\partial \mathbf{c}}\right) \\
& - \frac{\partial^2 \bar{W}}{\partial J_5 \partial J_5}\left(4\,\jmath^{5/3}/(\jmath_4\,\jmath_5^2)\mathsf{N}_{0c^{-1}}\,\frac{\partial \jmath_5}{\partial \mathbf{c}}\right) \\
& - \frac{\partial^2 \bar{W}}{\partial J_4 \partial J}\left(2\,\jmath/\jmath_4\mathbf{a}_0\,\mathbf{c}^{-1}\right) - \frac{\partial^2 \bar{W}}{\partial J_5 \partial J} \\
& \times \left(2\,\jmath^{5/3}/\jmath_4\mathsf{N}_{0c^{-1}}\,\mathbf{c}^{-1}\right) - \frac{\partial^2 \bar{W}}{\partial J^2}\,\jmath\mathbf{I}\mathbf{c}^{-1},
\end{aligned}
\tag{17}
$$

where $\mathbf{N}_{0c^{-1}} = \mathbf{a}_0 \cdot \mathbf{c}^{-1} + \mathbf{c}^{-1} \cdot \mathbf{a}_0$, $\mathsf{N}_{0c^{-1}} = \mathbf{a}_0 : \mathsf{I}_{c^{-1}} + \mathsf{I}_{c^{-1}} : \mathbf{a}_0$, while $\mathsf{I}_{c^{-1}} \equiv \partial \mathbf{c}^{-1} / \partial \mathbf{c}$, and I the symmetric identity 4th-order tensor. Also, the partial derivatives of the modified Eulerian invariants are given by $\partial \jmath_1 / \partial \mathbf{c} = \jmath^{-2/3}\, \mathbf{I} - \jmath_1\, 1/3\ \mathbf{c}^{-1}$, $\partial \jmath_2 / \partial \mathbf{c} = \jmath_1 \jmath^{-2/3}\, \mathbf{I} - \jmath^{-4/3}\, \mathbf{c} - \jmath_2\, 2/3\ \mathbf{c}^{-1}$, and $\partial \jmath_4 / \partial \mathbf{c} = \jmath^{-2/3}\, \mathbf{a}_0 \ - \jmath_4\, 1/3\ \mathbf{c}^{-1}$, $\partial \jmath_5 / \partial \mathbf{c} = \jmath^{-4/3}\, (\partial \imath_5 / \partial \mathbf{c}) - \jmath_5\, 2/3\, \mathbf{c}^{-1}$, since $\partial \imath_1 / \partial \mathbf{c} = \mathbf{I}$, $\partial \imath_2 / \partial \mathbf{c} = \imath_1\, \mathbf{I} - \mathbf{c}$, $\partial \jmath / \partial \mathbf{c} = \jmath\, 1/2\ \mathbf{c}^{-1}$, $\partial \imath_4 / \partial \mathbf{c} = \mathbf{a}_0$ and $\partial \imath_5 / \partial \mathbf{c} = \hat{\mathbf{n}}_0 \otimes (\mathbf{c} \cdot \hat{\mathbf{n}}_0) + (\hat{\mathbf{n}}_0 \cdot \mathbf{c}) \otimes \hat{\mathbf{n}}_0$.

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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